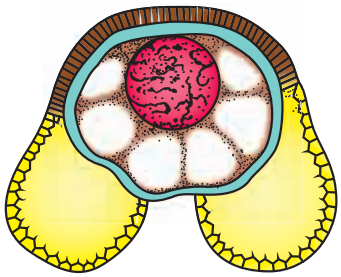
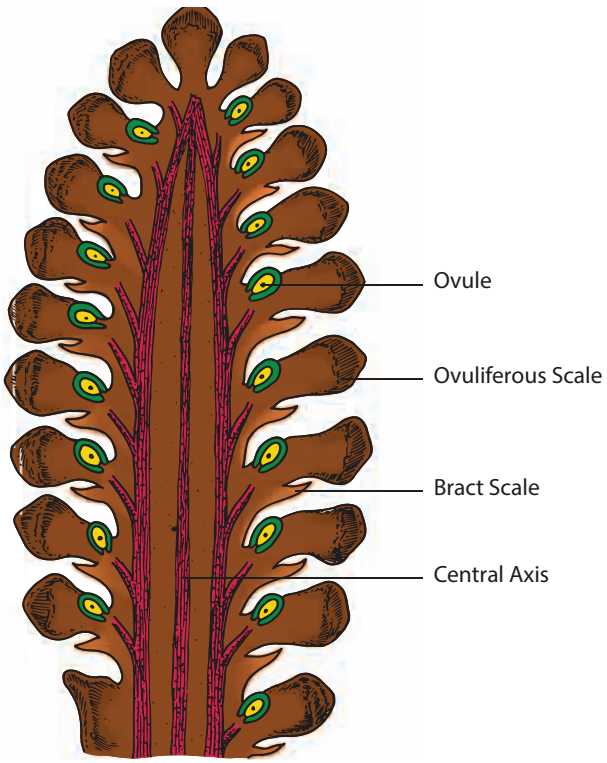
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c) A mature pollen grain



d) *Pinus* - A twig with female cone



e) L.S. of female cone

Figure 2.52: Reproduction in *Pinus*

Male cone

Male cones are produced in clusters on branches of unlimited growth. Each cone develops on the axil of scale leaf . The male cone consists of a centrally located cone axis surrounded by numerous spirally arranged microsporophyll. It bears two microsporangia at the base of the abaxial side of the microsporophyll. Each sporangium bears numerous winged microspores (n) or pollen grains. The microspores represent the male gametophyte

Female cone:-

Female cones are formed in the groups of 1 to 4 in the axils of the scale leaves. The female cone takes about three years to mature. It has the central axis around which megasporophylls are arranged spirally. The megasporophyll is the compound structure consisting of two types of scales. 1. Bract scale (sterile), and 2. Ovuliferous scales (fertile). The dorsal surface of each ovuliferous scale bears two ovules. Ovules bear megaspores which represent the female gametophyte.

Pollination and fertilization

In *Pinus* wind pollination takes place (Anemophilous). The microspore or pollen grain is released in the 4 celled stage(two prothallial cell, 1 generative and 1 tube cell). At the time of pollination a secretion oozes out from the micropyle of the ovule which entangles pollen grains which helps to lodge them in the pollen chamber. The tube cell protrudes to form pollen tube. The generative cell divides to produce stalk cell and body cell. The body cell divides into unequal male cells. Fertilization takes place after about a year of pollination. The pollen tube containing two male nuclei penetrates through the micropyle and reaches the egg. One of the male nuclei fuses with the egg forming diploid zygote and the remaning one gets degenerated. The fertilized egg